

OS Unit 2 Question Bank — CPU Scheduling & Synchronization

Operating Systems (OS) · Unit 2 · CPU Scheduling & Synchronization · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Write short notes on Scheduling Criteria.
2. Define FCFS, SJF and Round Robin. Derive or justify the key properties involved.
3. Discuss the role of Priority Scheduling in CPU Scheduling & Synchronization.
4. Compare Critical Section Problem with FCFS, SJF and Round Robin. When would you prefer one over the other?
5. Explain Semaphores with a neat diagram and a suitable example.
6. Compare Classic Synchronization Problems with FCFS, SJF and Round Robin. When would you prefer one over the other?

Part B — Long answer (10 marks)

1. Write short notes on Scheduling Criteria.
2. What is meant by FCFS, SJF and Round Robin? Discuss its advantages and limitations.
3. Illustrate Priority Scheduling with a worked example.
4. Write short notes on Critical Section Problem.
5. Define Semaphores. Derive or justify the key properties involved.
6. Discuss the role of Classic Synchronization Problems in CPU Scheduling & Synchronization.
7. Write short notes on Scheduling Criteria.
8. Compare FCFS, SJF and Round Robin with Scheduling Criteria. When would you prefer one over the other?
9. Explain Priority Scheduling with a neat diagram and a suitable example.
10. Write short notes on Critical Section Problem.
11. Define Semaphores. Derive or justify the key properties involved.
12. Define Classic Synchronization Problems. Derive or justify the key properties involved.